CS341 · COMPUTER GRAPHICS

Last-Minute Sheet

Definitions, comparisons and lists

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01 Definitions

Term	Definition
Computer graphics	The creation, storage, and manipulation of models and images.
Modeling · Rendering · Animation	Creating the geometry of objects · generating 2D images of them · describing how they change in time.
Resolution	The maximum number of points (pixels) that can be displayed without overlap on a CRT.
Persistence	The time it takes the emitted light to decay to one-tenth of its original intensity.
Refresh rate	The frequency at which a picture is redrawn on the screen.
Dot pitch	The space between the centers of adjacent pixels; for crisp images it should be less than 0.31 mm.
Aspect ratio	The ratio of vertical points to horizontal points needed to produce equal-length lines in both directions.
Frame buffer	The memory (refresh buffer, bit map) that stores the picture of a raster system; each location is one pixel.
Pixel · Scan line	A screen point (picture element) · one row of the raster.
Vector	Has magnitude and direction; no fixed location. Vector from P to Q: $v = Q - P$.
Unit vector	A vector of length 1: $\hat{a} = a \div \lvert a \rvert$ (normalizing).

Output primitives	Functions used to describe picture components: points, lines, circles, polygons, filled areas.
Coordinate extents	The min and max x, y, z of an object (bounding box).
DDA	Floating-point increments, then rounding. Drawbacks: rounding is time consuming, and pixels drift away from the true line on long segments.

02 Why? (reasons)

Question	Answer
Why use aspect ratio?	To maintain geometric features: a square stays a square and a circle does not become an ellipse.
Why are raster lines stair-stepped?	Each point is rounded to the nearest pixel, so all lines except horizontal and vertical look stair-stepped.
Why does random scan need little memory?	It stores only the display program (line-drawing commands).
Why are 99% of systems raster today?	Memory became cheap.
Why is LCD used more than plasma?	Plasma has better resolution but is more expensive.
Why is Point + Point not a point?	Its homogeneous coordinate is 1 + 1 = 2 (neither 0 nor 1).
Why low persistence for animation?	The light fades fast so motion does not smear; it needs a higher refresh rate.

03 Comparisons

	Random scan	Raster scan
Drawing	The beam draws lines directly	Sweeps the screen top to bottom, line by line
Storage	Display list	Frame buffer
Resolution	Very high, smooth lines	Stair-step lines
Color	Limited, no realistic shading	Easy color and shading
Memory	Little	A lot
Examples	Pen plotter, laser light show	PC monitors, TV, printers

	Active-matrix	Passive-matrix
Transistor	One per pixel	One per row or column
Quality	Brighter and sharper	Less sharp
Cost and power	More expensive, more complicated	Cheaper, needs less power

	Absolute	Relative
Meaning	The actual position, measured from the origin	An offset from the last position (current position)
Example	`(5, 7)`	`(3, 8) + (2, −1) → (5, 7)`

	Emissive	Nonemissive
Idea	Convert electrical energy into light	Use optical effects on sunlight or another light source
Examples	Plasma, LED, thin-film electroluminescent	LCD

	Point	Vector
Meaning	A location in space	Magnitude and direction, no location
Homogeneous	`(x, y, z, 1)`	`(x, y, z, 0)`

	Low persistence	High persistence
Suits	Animation (needs higher refresh)	Complex static pictures

	Line	Segment	Ray
t	All t	$0 \leq t \leq 1$	$t \geq 0$

04 Lists

List	Answer
Input devices	Keyboard · Mouse · Trackball · Joystick · Data glove · Digitizer · Scanner · Touch panel · Light pen · Voice system
Hard-copy devices	Laser printer · Ink-jet printer · Film recorder · Electrostatic printer · Pen plotter
Video display devices	CRT (refresh CRT, DVST, color CRT) · LCD · Plasma panels · Thin-film electroluminescent
Applications of CG	CAD · Medical visualization · Entertainment · Simulation · Education · Training · Scientific visualization · User interfaces
Main tasks	Modeling · Rendering · Animation
Screen clarity depends on	Resolution · Dot pitch · Refresh rate
Flat-panel advantages	Thinner · Less weight · Less power · No radiation
CRT parts	Electron gun · Focusing system · Deflection plates · Phosphor-coated screen · Connector pins
Color CRT methods	Beam penetration · Shadow mask
OpenGL libraries	OpenGL core (gl) · GLU (glu) · GLUT (glut)
Random scan disadvantages	Needs an intelligent beam · flicker limit · can't draw complex images · limited color · no realistic shading

05 Problems and rules

Aspect ratio: $AR = \frac{\frac{Yp}{Ymax}}{\frac{Xp}{Xmax}}$ resolution is written width × height

DDA: steps = max(|dx|, |dy|) $xInc = \frac{dx}{steps}$ $yInc = \frac{dy}{steps}$

round(a) = int(a + 0.5)

|m| < 1: x + 1, y + m |m| > 1: y + 1, $x + \frac{1}{m}$

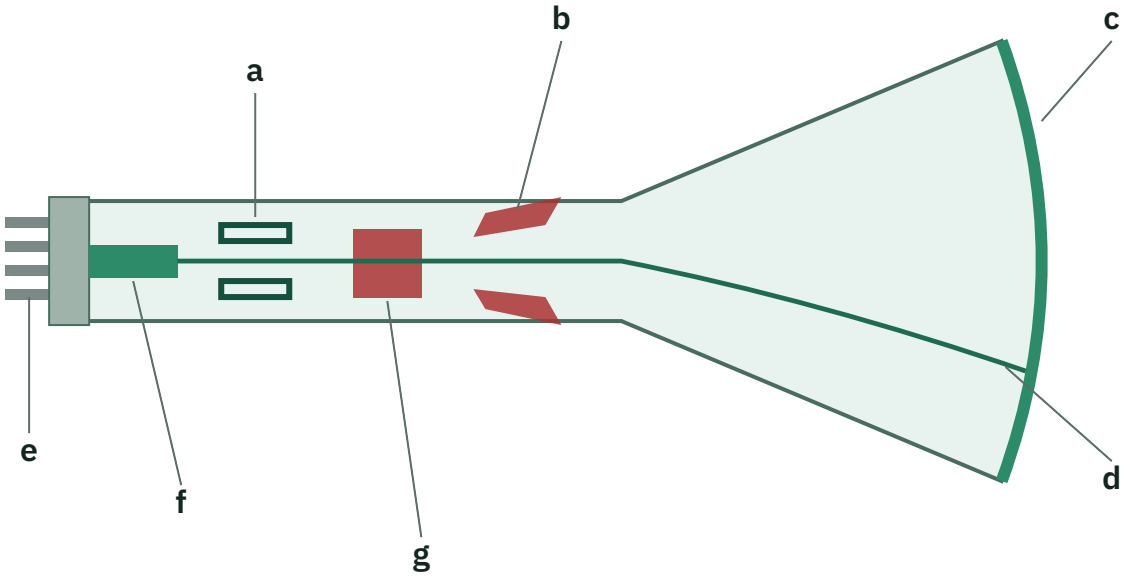
GL_LINES: every 2 points = a line (an odd point is ignored)
GL_LINE_STRIP: connected GL_LINE_LOOP: connected + closed

GL_TRIANGLES: 3 vertices per triangle Point − Point = Vector
Point + Vector = Point $a \cdot b = 0 \rightarrow$ perpendicular

06 The CRT figure

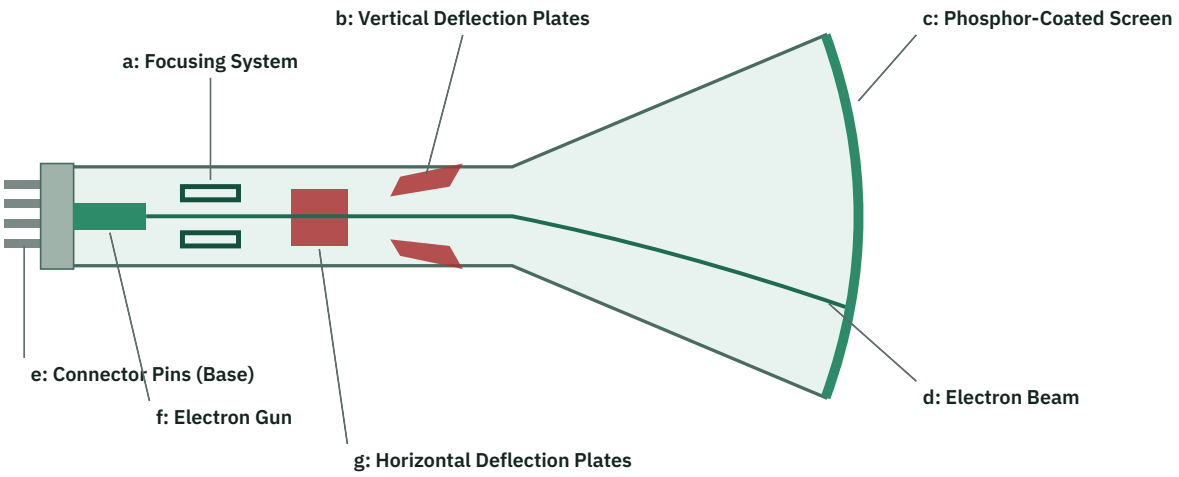
This is how it comes in the midterm (472): the figure with letters, and you write the name of each letter. Practice: cover the answer and fill it in:

Label the cathode ray tube using the following letter components:



Letter	Answer	Letter	Answer
a		e	
b		f	
c		g	
d			

The answer



Letter	Part	What it does
a	Focusing System	Focuses the beam to a point, like a lens
b	Vertical Deflection Plates	Move the beam up and down
c	Phosphor-Coated Screen	Glows when an electron hits it
d	Electron Beam	The beam itself
e	Connector Pins (Base)	Connect the tube to the circuit
f	Electron Gun	Fires the electrons
g	Horizontal Deflection Plates	Move the beam left and right

From the left: connector pins → electron gun → focusing system → horizontal plates → vertical plates → phosphor screen.

Memorize it in the order the electron travels from left to right: it leaves the gun, is focused, is deflected by the plates, and hits the screen.